

HAOHAI SHI

Department of Physics

University of Maryland, College Park

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EDUCATION

University of Maryland, College Park

Ph. D. in Physics, GPA 3.90/4.00

Chu Kochen Honors College, Zhejiang University

BSc in Physics

Maryland

Sep 2022 – Present

Hangzhou, China

Sep 2018 – Jun 2022

- GPA: 3.95/4.00 (91.2/100; rank top 5%)
- Bronze Medal of 34th Chinese Physics Olympiad (CPhO)

PUBLICATIONS & MANUSCRIPTS

- Yao, Y, *et al.* "Observation of Many-Body Fock Space Dynamics in Two Dimensions." *Nature Physics* 19, no. 10 (2023): 1459–65. <https://doi.org/10.1038/s41567-023-02133-0>
- Shi, H. *et al.*, An Error Suppressing Encoding for Positional Disorder in Rydberg Atom Analog Quantum Simulation, Manuscript in preparation (2026)

RESEARCH EXPERIENCE

University of Maryland, College Park (QuICS)

Research Assistant to Professor Xiaodi Wu

College Park, Maryland

Jun 2023 – Present

NAP-Sim: Neutral-atom Array Performance Simulator for Throughput-Aware Architecture

- A discrete-event simulator (DES) that represents tweezer array operations as capacity-constrained resources with precedence, enabling design-space exploration under realistic timing, contention, and scheduling.
- Use the DES to quantify marginal improvements on shot wall time and logical-error rates. Verify or challenge prevailing beliefs, reveal trade-offs, and outline Pareto fronts that guide hardware–software co-design.
- Provides a transparent, reproducible framework for architectural decision-making, turning rules of thumb into testable, quantitatively supported choices.

RISC-Q: A Generator for Real-Time Quantum Control System-on-Chips Compatible with RISC-V

- A programming interface compatible with the RISC-V ecosystem, developed with SpinalHDL. Enabling efficient automation of highly parameterized and modular QCSoc architectures.
- Ongoing: Implementing a pipelined hardware decoder for lower latency and compatibility with general qLDPC codes.

Error Suppressing Encoding for Positional Disorder in Rydberg Atom Analog Quantum Simulation

- An experiment-ready encoding for Rydberg atom arrays that simulates a family of XXZ Heisenberg models.
- Construct a "super atom" encoding, where each logical atom is represented by m physical atoms within the Rydberg blockade regime.
- Theoretically predict a reduction of error by a factor of $1/\sqrt{m}$ and verify via numerical simulation.

Quantum Phase Transitions in non-Hermitian Transverse Field Ising Model (NH-TFIM)

- Observe a sharp change of magnetization when tuning the strength of imaginary fields at zero temperature for 1D Ising chain.
- Analytical expansion solutions and large-scale solution benchmark with biorthonormal block (bb)-DMRG and Time-Evolving Block Decimation (TEBD).
- Implement non-unitary dynamics on IonQ devices using linear combination of Hamiltonian simulation (LCHS) and Schrodingerization.

Zhejiang University (Provincial Key Lab for Quantum Technologies and Devices)

Research Assistant to Professor Haohua Wang, Assistant Dean of the Physics Department

Hangzhou, China

Aug 2021 – Jun 2022

Microwave Package Design for Superconducting Chip

- Assembly, characterization, calibration and control of a 36-qubit superconducting quantum processor.
- Designed a scalable 3D 24-qubit package, significantly reduced signal crosstalk via HFSS FEM analysis.

North Carolina State University (Department of Electrical and Computer Engineering)

Undergraduate Summer Research Assistant to Professor Huiyang Zhou

Online

Jul 2021 – Aug 2021

Quantum Device Design and Analysis Based on Qiskit Metal

- Analyzed the key properties of the qubits, waveguides, and resonators. e.g. eigenmodes, impedance, and energy participation ratio (EPR) using qiskit metal.

Zhejiang University (Center for Correlated Matter)

Research Assistant to Professor Xin Lu,

Hangzhou, China

Sept 2020 – Jun 2021

Scanning Probe and Torque Differential Magnetometry Based on Quartz Crystal Oscillator

- Developed a prototype scanning probe with sub-micron-level resolution, integrated into point-contact spectroscopy (PCS); Implemented Phase-Lock Loop (PLL) scheme, successfully observed the de Haas van Alphen (dHvA) quantum oscillation in heavy-fermion superconductor CeRnIn5.

ADDITIONAL INFORMATION

Coding and Language Skills

- Efficient toolkits: iTensor, Stim, QuTiP, Qiskit (Aer, Metal), Bloqade, dynamite, Mathematica, SimuQ,

Languages

- English (fluent), Chinese (native), Cantonese (fluent)

Interests

Soccer, ski, bouldering, geek crafts.

Member of the Soccer Team CSSA@UMD (Aug. 2022-Present), champion of UMD intramural (graduate level).

- DIY open source game project:
Tron, an online flight game inspired by movie Tron, implemented on WebGL Javascript.
(<https://github.com/ShawHaines/Tron>)